Programming Worlds

PROGRAMMING WORLDS ACADEMY

Git & GitHub 101

From Zero to Your First Pull Request

BEYOND CODE. BEYOND LIMITS.

05

MODULE 05

Connecting Git to GitHub

Link your local repository to GitHub and publish your first project online.

MODULE 05 · LESSON 5.1

Local vs remote repositories

المستودع المحلي مقابل البعيد

Learning objectives

- Explain what a GitHub repository is.
- Read a remote URL.
- Explain what `origin` means.
- Tell `main` from `origin/main`.
- Tell HTTPS from SSH URLs.

Prerequisites

Module 4 and your GitHub account.

Introduction

So far your history lives only on your computer. GitHub holds a second copy online. This lesson explains how the two copies relate before you connect them.

Explanation

A **GitHub repository** is the remote copy of your project. It stores the same commits you have locally, and adds a web page, sharing and teamwork tools.

A **remote** is a saved bookmark to another repository. Its **URL** says where it lives. The HTTPS form is `https://github.com/USER/REPO.git`. The SSH form is `git@github.com:USER/REPO.git`.

origin is only a name. By convention Git calls the main remote `origin`. You could name it differently, but everyone uses origin.

Branches. `main` is your local branch. `origin/main` is your computer's remembered copy of the remote's main branch (a *remote-tracking branch*). It updates only when you push, fetch or pull.

HTTPS or SSH? HTTPS is simpler for beginners: Git Credential Manager signs you in through the browser. SSH uses a key pair and no password prompts after setup. This course uses HTTPS.

Real-world analogy

> Your local repo is your notebook. GitHub is the photocopy kept at the library. `origin` is the library's name on your address list.

Diagram

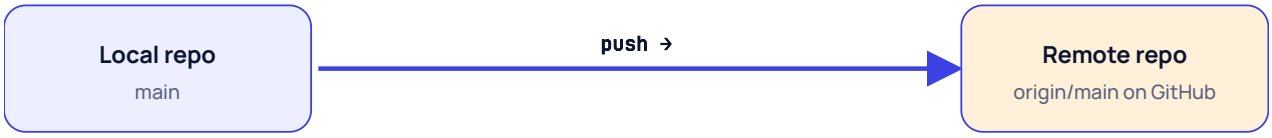


Figure 1 · Two copies of the same history

Commands

No new commands yet. Lesson 5.2 uses `git remote` and `git push`. Figure 1 shows how the two copies exchange commits.

Common mistakes and solutions

Thinking GitHub updates by itself.	GitHub only changes when you push.
Thinking origin is a special keyword.	It is a conventional name for a remote.
Copying the wrong URL type.	Use the HTTPS URL if you chose HTTPS.

Practical exercise

> Open your GitHub profile. Click New to start a repository and look at the options, but do not create it yet. Find the Public and Private choices.

Expected result

You can point to the repository name box and the visibility choice.

Knowledge check

1. What is a remote?

- ○ A commit message
- ○ A saved link to another repository
- ○ A deleted branch

2. What does origin mean?

- ○ A conventional name for the main remote
- ○ The first commit
- ○ Your username

3. What is origin/main?

- ○ A GitHub setting
- ○ Your computer's copy of the remote main branch
- ○ A new file

Answers

- A saved link to another repository — A remote stores the URL of another repository.
- A conventional name for the main remote — It is just a name.
- Your computer's copy of the remote main branch — It is a remote-tracking branch.

Summary

GitHub holds the remote copy. A remote is a named URL, usually origin. main is your branch; origin/main is the remembered remote one.

Key vocabulary

Remote	A named link to another repository
origin	Conventional name of the main remote
Remote-tracking branch	Local record of a remote branch, such as origin/main

Next lesson

Lesson 5.2 — git remote and git push. Now you connect and upload.

MODULE 05 · LESSON 5.2

git remote and git push

git remote ๑git push

Learning objectives

- Create an empty repository on GitHub.
- Connect it with `git remote add`.
- Push with `git push -u origin main`.
- Explain `-u` and upstream.
- Fix common push errors.

Prerequisites

Lesson 5.1 and at least one commit.

Introduction

This is the step where your work first appears online.

Explanation

Create the repository. On GitHub choose **New**. Name it `my-first-project`. Leave "Add a README" and other files **unchecked**, so the repository starts empty. Create it and copy the HTTPS URL.

`git remote add origin <URL>` saves the remote under the name origin. `git remote -v` lists remotes (Figure 1).

`git branch -M main` renames your current branch to `main`. It is safe, and ensures the name matches GitHub.

`git push -u origin main` uploads the `main` branch to `origin`. The **-u** option sets the **upstream**: it tells Git that local `main` goes with `origin/main`. After that, plain `git push` is enough.

First vs later pushes. The first push needs `-u origin main`. Later pushes are just `git push`.

Authentication. On the first push a Git Credential Manager window opens. Sign in to GitHub in the browser. Git remembers it afterwards.

Confirm. Refresh the repository page: your files and commits are there.

Real-world analogy

> `git remote add` saves the library's address. `git push` is posting the parcel. `-u` remembers the address so next time you just say "post it".

Diagram

Illustration · Git Bash (not a real screenshot)

```
$ git branch -M main
$ git push -u origin main
To https://github.com/YOUR-USERNAME/my-first-project.git
 * [new branch]      main → main
branch 'main' set up to track 'origin/main'.
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Figure 1 · Illustration of the expected output

Commands

`git remote add origin https://github.com/YOUR-USERNAME/my-first-project.git`
`git remote -v`
`git branch -M main`
`git push -u origin main`
`git push`

Options: `-v` verbose list; `-M` force-rename the branch; `-u` set upstream. Replace YOUR-USERNAME with yours.

Common mistakes and solutions

“remote origin already exists”	Use `git remote set-url origin <URL>` to change it.
“Repository not found”	Check the URL spelling and that you are signed in to the right account.
“Authentication failed”	Sign in again through the browser window. Do not use your password in the terminal.
“rejected … non-fast-forward”	The remote has commits you do not have. You will fix this with pull in Module 8.

Practical exercise

> Create the empty GitHub repository, add the remote, and push `main`.

Expected result

Git prints “branch ‘main’ set up to track ‘origin/main’” and GitHub shows your three commits and three files.

Knowledge check

1. What does -u do in git push -u?

- ○ Updates Git
- ○ Sets the upstream branch
- ○ Uploads faster

2. Which command lists remotes?

- ○ git remote -v
- ○ git push -v
- ○ git list

3. Do commits reach GitHub with git commit?

- ○ Only on Mondays
- ○ No, you must push
- ○ Yes, automatically

Answers

- Sets the upstream branch — It links local main with origin/main.
- git remote -v — git remote -v shows names and URLs.
- No, you must push — Commit is local; push uploads.

Summary

You created an empty GitHub repository, linked it as origin, and pushed main with -u. Later you only need git push.

Key vocabulary

Upstream	The remote branch your local branch tracks
git push	Upload commits to a remote
-u	Set the upstream while pushing

Next lesson

Lesson 5.3 — Exploring a GitHub repository. See what your push looks like online.

MODULE 05 · LESSON 5.3

Exploring a GitHub repository

استكشاف مستودع GitHub

Learning objectives

- Find files, commits and branches on GitHub.
- Read the README on the page.
- Name Issues and Pull requests.
- Find visibility and URLs.

Prerequisites

Lesson 5.2.

Introduction

Your pushed project now has a web page. Learn its parts, because you will use them every day.

Explanation

Code tab: files and folders, the branch selector (`main`), and the green **Code** button with the clone URL.

The **README** shows below the file list, formatted. It is the front page of the project.

Commits: click the clock icon with the commit count to see every commit with message, author and date.

Branches: lists all branches. You will create more in Module 7.

Issues track tasks and bugs. **Pull requests** propose changes for review (Module 9).

Settings controls visibility (Public or Private), collaborators, and later GitHub Pages. Change visibility carefully.

URLs. The page address is `https://github.com/USER/REPO` . The `.git` version of it is what Git uses to clone and push.

Real-world analogy

> The repository page is a shop window for your project: the front page (README), the stock list (files), the history book (commits).

Diagram

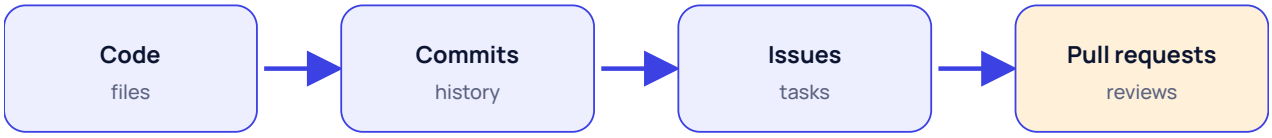


Figure 1 · Main tabs of a GitHub repository

Commands

No Git commands here. You work in the browser.

Common mistakes and solutions

Making a repo public by accident.	Check the visibility under Settings.
Editing files on GitHub then forgetting local.	Later you must pull those changes (Module 8).
Ignoring the README.	Write it early. It is the first thing people read.

Practical exercise

> Open your repository on GitHub. Find: the file list, the commit count, the branch selector, the README and the Settings tab.

Expected result

You can locate all five and open the commit list showing your three commits.

Knowledge check

1. Where do you see commit history?

- ○ Settings
- ○ The Commits link on the Code tab
- ○ The Issues tab

2. What are Pull requests for?

- ○ Proposing changes for review
- ○ Deleting repos
- ○ Installing Git

3. Where is visibility set?

- ○ README
- ○ Settings
- ○ Issues

Answers

- The Commits link on the Code tab — The commit count opens the full history.
- Proposing changes for review — They are how changes are reviewed and merged.
- Settings — Settings holds Public and Private.

Summary

A GitHub repository page shows code, commits, branches, README, issues, pull requests and settings, including visibility.

Key vocabulary

README	The project's front page
Issue	A tracked task or bug
Pull request	A proposal to merge changes

Next lesson

Module 6 — Git and GitHub in VS Code. You will do all of this with buttons.

MODULE 05

Practical lab 5 · Push your first project to GitHub

1. Create an empty repository named `my-first-project` on GitHub.
2. In your project folder run `git remote add origin <your URL>`.
3. Run `git remote -v`.
4. Run `git branch -M main` and `git push -u origin main`. Sign in when asked.
5. Refresh GitHub and open the commit list.

Check	Pass if…
Remote	`git remote -v` shows origin with your URL
Upload	GitHub shows your 3 files and 3 commits
Upstream	`git status` says the branch is up to date with `origin/main`

Evaluation: 3 of 3 checks = pass.

Module 5 quiz (pass mark: 70%)

1. Which command links a remote named origin?

- ○ git push origin
- ○ git clone origin
- ○ git remote add origin URL

2. What does git push -u origin main set?

- ○ Your email
- ○ The upstream branch
- ○ The commit message

3. What does origin usually mean?

- ○ The main remote
- ○ The first file
- ○ A branch

4. What does main versus origin/main show?

- ○ Two users
- ○ Local branch versus remembered remote branch
- ○ Two folders

5. Which URL type does this course use?

- ○ FTP
- ○ Bluetooth
- ○ HTTPS

Answers

- git remote add origin URL — remote add saves the URL under a name.
- The upstream branch — It tracks origin/main.
- The main remote — It is the conventional remote name.
- Local branch versus remembered remote branch — origin/main is a remote-tracking branch.
- HTTPS — HTTPS with Git Credential Manager.